Title Page

Manuscript title:
Counterfactual Attribution of Crypto Systemic Stress on an Auditable On-Chain Graph

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Author's Information

Quang-Vinh Dang. Quang-Vinh Dang is a Lecturer at British University Vietnam and Data/AI Director at SmartOSC, where he leads initiatives in multi-agent AI systems, large language models, and graph neural networks across banking, retail, and finance. He holds a Ph.D. in Computer Science from Université de Lorraine (LORIA), France (2018), an M.Sc. in Financial Engineering from WorldQuant University (2021), and completed a Post-Master programme in Software Technology at Eindhoven University of Technology. His research spans machine learning in finance, natural language processing, intrusion detection, and federated learning, and he has authored more than 60 peer-reviewed papers indexed in Scopus/SCI-E, many in Q1 journals. Prior to his current roles, he held data-leadership positions at Trusting Social, Be Group, GoTymeX, and DatVietVAC, and was a postdoctoral researcher at Inria, France.

Cover Letter

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The Editor-in-Chief
Digital Finance

Dear Editor-in-Chief,

I am submitting the manuscript **"Counterfactual Attribution of Crypto Systemic Stress on an Auditable On-Chain Graph"** for consideration in *Digital Finance*. Crypto-asset markets have repeatedly transmitted distress between exchanges, lending protocols, and stablecoin issuers,

most recently in the liquidation cascade of October 2025. Measures of this contagion commonly infer the network from price co-movement, using CoVaR-type or marginal expected shortfall statistics. The resulting graphs are estimated rather than observed, and the attributions built on them cannot be checked against the transfers that moved funds between entities. Public blockchains record those transfers, but the raw ledger becomes an entity-level graph only after addresses are resolved to institutions.

The manuscript treats the entity-resolved on-chain ledger as an auditable candidate skeleton for a linear structural causal model of entity stress. On this model it defines On-Chain Causal CoVaR and a counterfactual loss attribution. The analytic results are central. The interventional effect of one entity's stress on another entity's tail measure is a network multiplier that depends on the lagged coefficients and the directed paths, not on the flows or the shock distribution. This yields flow-free bounds and implies zero attribution between entities with no directed path. The attribution is invariant to how event-window shocks are abducted, so that choice affects only the hinge loss. The linear Shapley attribution has a closed-form walk decomposition, and an exact subset algorithm computes it with 2^p rather than $p! \cdot p$ model evaluations. The paper gives equal weight to the auditable ledger-based attribution and to its application to the October 2025 cascade, with checks on the FTX (2022) and USDC/SVB (2023) episodes. In that application the hourly exported-risk ranking differs from ΔCoVaR (Spearman 0.52, six entities with matched price series). A deterministic zero-shock proxy of the ranking is stable under coefficient perturbation (mean Spearman 0.88 over 25 draws), which checks the method rather than the reported ranking. The analytic checks refer to the daily specification. The advantage of the observed graph over a price-estimated graph at hourly frequency is small and not consistent across specifications, and I report it as weak evidence.

The paper fits the scope of *Digital Finance* on three counts. It addresses systemic risk in crypto-asset markets and how that risk is measured. It addresses supervisory transparency, since each attribution traces back to public ledger records and an explicit model. It also provides reproducible inputs: the code, the query definitions, and the derived result files are openly available. Raw query outputs are regenerated from public Dune Analytics data.

This manuscript is original, has not been published previously, and is not under consideration at any other journal. It is single-authored, and the author declares no competing interests. The code and derived results are publicly available at https://github.com/vinhqdang/VEIN-On-Chain-Causal-Risk-Attribution-for-Cryptocurrency-Systemic-Risk. Thank you for considering this manuscript.

Sincerely,

Quang-Vinh Dang
British University Vietnam

Declarations

Ethical Approval and Consent to participate. Not applicable: this study uses only public,

aggregated on-chain and market data and involves no human subjects, human data, or animals.

Consent for publication. Not applicable.

Availability of supporting data. The complete implementation, the real-data evaluation pipeline, and all result artefacts underlying every number, table, and figure in this paper are publicly available at https://github.com/vinhqdang/VEIN-On-Chain-Causal-Risk-Attribution-for-Cryptocurrency-Systemic-Risk. The repository includes the full source code, the SQL issued to Dune, and the scripts and derived result files that reproduce every reported result. All underlying blockchain and market data are public and were obtained from Dune Analytics (https://dune.com), DefiLlama (https://defillama.com), and Etherscan (https://etherscan.io); no restricted or proprietary data were used.

Competing interests. The author declares no competing interests.

Funding. No external funding supported this work.

Authors' contributions. Q.-V.D. conceived the study, developed the VEIN framework and its implementation, performed the data collection and empirical analysis, and wrote the manuscript.

Acknowledgements. Not applicable.

Authors' information (Optional). See "Author's Information" above.